FILIPPO MARCANTONI

Robotics Software Engineer | Computer Vision | Robot Learning | Physical AI
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SUMMARY

Robotics engineer with an M.S. in Robotics Engineering from Worcester Polytechnic Institute and R&D experience across computer vision, robot learning, industrial automation, generative AI, swarm robotics, and teleoperation. Experienced in developing full-stack robotic systems that integrate perception, learning, simulation, control, and deployment under real-world constraints. Seeking robotics software, autonomy, robot learning, computer vision, or physical AI roles.

EDUCATION

Worcester Polytechnic Institute (WPI) — *Worcester, MA, USA*
M.S. Robotics Engineering | GPA 3.90/4.00 — *Aug 2024 – May 2026*
B.S. Robotics Engineering | GPA 3.89/4.00, Summa Cum Laude — *Aug 2021 – May 2025*

EXPERIENCE

AI & Industrial Robotics R&D Intern | B&R Industrial Automation, Austria — *Jun 2025 – Aug 2025*
- Developed an LLM-based code-generation copilot for B&R Automation Studio using Transformers, RAG, and Azure AI retrieval to support industrial robot programming and automation workflows.
- Designed a robotics-specific retrieval backend with domain-aware metadata extraction and syntax-aware chunking for Structured Text and Motion ST code, improving context quality across large automation codebases.

Machine Vision & AI R&D Intern | Makro Labelling SRL, Italy — *May 2024 – Aug 2024*
- Developed C++ and Python machine-vision pipelines for bottle orientation, label alignment, and defect inspection on high-speed labeling lines.
- Benchmarked classical and deep-learning models across bottle formats, lighting, and label shifts to identify failure modes and improve automated quality-control robustness.

RESEARCH

STL-to-Fluoroscopy 2D–3D Registration | FuTURE Lab, WPI — *Aug 2025 – Dec 2025*
- Designed a self-supervised fluoroscopy-to-volume registration workflow using DiffDRR, 3D Slicer, and DICOM processing for image-guided endovascular procedures.
- Validated patient-specific pose estimation on a 3D-printed aorta phantom without ground-truth labels.

VR-Based Multimodal Teleoperation Interface | HiRo Lab, WPI — *Jan 2025 – May 2025*
- Built a VR teleoperation system for a robotic nursing assistant using Unity, ROS, Meta Quest head-motion mirroring, real-time multi-camera streaming, Whisper voice commands, and ZED-Mini 3D object detection.
- Integrated viewpoint control, speech-based camera switching, and perception-driven camera selection to improve operator situational awareness during remote manipulation.

PiELo: Reactive Swarm Programming Language | NEST Lab, WPI — *Aug 2024 – May 2025*
- Designed PiELo, a C++ reactive swarm-programming language with consensus and virtual stigmergy as first-class primitives for distributed robotic coordination.
- Validated distributed swarm behaviors in ARGoS3 and on Khepera IV robots with Vicon motion tracking.

SELECTED PROJECTS

Robot Learning: Reinforcement & Imitation Learning
- Implemented policy iteration, REINFORCE, A2C, and A3C for LunarLander and PyBullet Kuka vision-based grasping, reaching 55% grasp success with a CNN actor-critic policy.
- Trained robomimic policies on 60 PickPlaceCan demonstrations, improving success from 0.02 with baseline behavioral cloning to 0.80 with LSTM behavioral cloning and 0.90 with diffusion policy.

Classical MSCKF & Deep Visual-Inertial Odometry
- Implemented a stereo MSCKF VIO backend with IMU propagation, camera-state cloning, stereo feature tracking, and nullspace EKF updates, evaluated on EuRoC and RPG datasets.
- Trained IMU-only, vision-only, and fused deep VIO networks on synthetic Blender trajectories using geodesic SO(3) and trajectory-composition losses.

Monocular 3D Driving-Scene Reconstruction
- Built a monocular autonomous-driving perception pipeline integrating 3D object detection, segmentation, depth estimation, pose estimation, and optical flow.
- Reconstructed 3D driving scenes in Blender with lanes, vehicles, pedestrians, traffic objects, orientation, motion state, and collision-risk estimation.

3D Scene Reconstruction: Classical SfM & NeRF
- Implemented Structure-from-Motion from scratch, including SIFT tracking, RANSAC, fundamental matrix estimation, essential-matrix recovery, cheirality checks, PnP-RANSAC, triangulation, and bundle adjustment.
- Built a PyTorch NeRF with positional encoding, 5D MLPs, hierarchical sampling, and differentiable volume rendering.

TECHNICAL SKILLS

Languages: Python, C++, C, C#, MATLAB, Java, JavaScript, Buzz.
Robotics & Simulation: ROS/ROS2, Gazebo, MuJoCo, PyBullet, ARGoS3, robosuite, robomimic, Unity, Blender, Vicon, VR/XR, Dynamixel, ZED SDK, Automation Studio, 3D Slicer.
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, TensorRT, CUDA, Transformers, LLMs, RAG, Reinforcement Learning, Imitation Learning, Behavioral Cloning, Diffusion Policy, Actor-Critic Methods.
Computer Vision & 3D Perception: OpenCV, SLAM, Visual-Inertial Odometry, MSCKF, SfM, NeRF, RANSAC, PnP, Bundle Adjustment, Homography, Object Detection, Segmentation, Monocular Depth, Pose Estimation, Optical Flow, Point Clouds.
Tools & Infrastructure: Git, Docker, Linux, Jupyter, Azure AI, AWS, Simulink, React, Node.js, PostgreSQL, MySQL, HPC/Slurm.
CAD & Design: SolidWorks, Fusion 360.